removed during replication of DNA

b. Segment of a chain of nucleotides formed during replication of DNA

c. Segments of gene which undergo recombination

d. Segments of DNA capable of replication

13. DNA replication is aided by
    a. DNA Polymerase only
    b. Both DNA polymerase and primase only
    c. DNA ligase only
    d. RNA Polymerase.

14. The semi-conservative mode of DNA replication was proved by
    a. Beadle and Tatum
    b. Meselson and Stahl
    c. Watson and Crick
    d. H.G Khorana

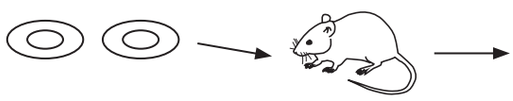
15. Enzymes involved in unwinding of DNA at replication are
    a. Ligases
    b. Helicases
    c. Endonucleases
    d. DNA Polymerases

16. DNA replication is semiconservative because the _____ strand will become half of the _____molecule
    a. RNA, DNA
    b. Template, finished
    c. Sense, mRNA
    d. Condon, anticodon

17. In DNA adenine is complementary base for _____ and cytosine is the complementary for _____
    a. Guanine, thymine
    b. Uracil, guanine
    c. Thymine, guanine
    d. Thymine, uracil

Answer the following

1. Define gene
2. DNA is not always the genetic material, what are the exceptions?
3. Define Nucleotide.
4. List any two difference between DNA and RNA
5. In what sense are the two strands of DNA antiparallel.
6. What is a nucleoside?
7. Depict Erwin Chargaff rule by an equation.
8. Give examples of nitrogenous bases.
9. Draw the structure of Deoxyribose.
10. State Watson and Crick rule.
11. List 2 characteristics of Z DNA
12. Define replication.
13. What is a template DNA?
14. Explain **semiconservative** replication.
15. List major events in replication
16. Write two main events during initiation of replication.
17. What is the role of topoisomerase?
18. What do you understand by leading strand/lagging strand?
19. What is RNA primer?
20. Name two enzymes that make primers for DNA synthesis
21. What is the origin of replication?
22. Label the following diagram of Griffith.



23. Point out the mistake in the following scheme:

    Extracts of smooth strain + R**Nase**

    ⟶ Mouse ⟶ Alive

24. Differentiate between right handed and left handed DNA forms with any three salient features.

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